



$$\text{Power Scale} = \sqrt{3} \times V_{\text{rated}} \times \text{current scale} = \text{_____} W = 1 \text{ cm}$$

PERFORMANCE AT FULL LOAD FROM CIRCLE DIAGRAM

$$\text{Full load output} = \text{_____} W = PQ = \text{_____} \text{ cm}$$

$$\text{Full load current} = OP \times \text{current scale} = \text{_____} \times \text{_____} = \text{_____} A$$

$$\text{Full load power factor} = \frac{PT}{OP} = \text{_____} \text{ lag}$$

$$\text{Rotor copper loss at full load} = QR \times \text{power scale} = \text{_____} \times \text{_____} = \text{_____} W$$

$$\text{Stator copper loss at full load} = RS \times \text{power scale} = \text{_____} \times \text{_____} = \text{_____} W$$

$$\text{Constant loss} = ST \times \text{power scale} = \text{_____} \times \text{_____} = \text{_____} W$$

$$\text{Rotor input at full load} = PR \times \text{power scale} = \text{_____} \times \text{_____} = \text{_____} W$$

$$\text{Torque developed at full load} = PR \times \text{power scale (sync. watts)} = PR \times \text{power scale} \times \frac{60}{2\pi N_s} \text{ N-m}$$

$$= \text{_____} \times \text{_____} \times \frac{60}{2\pi \times N_s} = \text{_____} \text{ N-m}$$

$$\text{Stator input at full load} = PT \times \text{power scale} = \text{_____} \times \text{_____} = \text{_____} W$$

$$\text{Efficiency at full load} = \frac{PQ}{PT} \times 100\% = \text{_____}\%$$

$$\text{Slip at full load, } s = \frac{QR}{PR} \times 100\% = \text{_____}\%$$

$$\text{Speed at full load} = (1-s) \times N_s = \text{_____} \text{ rpm}$$

$$\text{Starting torque developed} = BG \times \text{power scale} \times \frac{60}{2\pi N_s} \text{ N-m}$$

$$= \text{_____} \times \text{_____} \times \frac{60}{2\pi \times N_s} = \text{_____} \text{ N-m}$$

To find the maximum torque, draw the perpendicular to torque line from centre C to meet the circle at I and the vertical I I' represents maximum torque to scale.

$$\text{Maximum torque} = I I' = \text{_____} \times \text{_____} \times \frac{60}{2\pi \times N_s} = \text{_____} \text{ N-m}$$

To find the maximum output, draw the perpendicular to output line from centre C to meet the circle at H and the vertical H H' represents maximum output to scale.

$$\text{Maximum output} = H H' = \text{_____} \times \text{_____} = \text{_____} W$$

EQUIVALENT CIRCUIT PARAMETERS